



Trash to treasure: Gamification and informed recycling behavior

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ABSTRACT

Enhancing sustainability entails promoting recycling in consumer education. Interactive digital platforms, websites, and apps that offer insights and incorporate gamification to interest and incentivize users represent an emerging trend. These include interactive edutainment and recycling games, tailored recycling schemes, and virtual and tangible incentives for specific behaviors. This study is based on 26 interviews and an app usage experiment with 18 participants to evaluate the effects of different information provision (paper-based vs. app and gamified app) modes to engage participants in waste management. The study considers behavior modification by examining the potential effect of gamification on informed recycling. By adopting determinants from behavioral psychology (e.g., social norms, peer influence, and positive reinforcement), this paper focuses on sustainable recycling behavior. It contributes novel perspectives by identifying recycling-oriented games, refining existing theoretical recycling models, and suggesting effective approaches for policymakers and businesses.

1. Introduction

Household participation is crucial for effective waste management (Kaplan Mintz et al., 2019; Kushwah et al., 2023; Salem et al., 2020; Zhang et al., 2023). Despite government efforts to enact regulations and standards, separating household waste is still in its early stages, and a gap exists between promoting eco-friendly behaviors and providing recycling guidance (Chan, 2022; Liu & Mukheibir, 2018). By consolidating previous research, Knickmeyer (2020) shows that educating households on the necessity and advantages of waste separation and teaching them how to separate waste fractions to meet the needs and seize the opportunities of the local waste management system are the keys to transitioning toward a circular economy. Individuals may not fully implement their intentions, resulting in an intention–behavior gap (Echegaray & Hansstein, 2017; Kushwah et al., 2023; Soma et al., 2021; Wang & Mangmeechai, 2021). Bridging this gap is crucial for both theory and practice. This study draws inspiration from the causality flow outlined in the theory of planned behavior (TPB) to critically examine the relevance of attitudes, subjective norms, perceived behavioral control, and intentions to achieve effective waste separation decisions (Chen & Gao, 2021; Hu et al., 2024). However, it diverges from

conventional TPB studies in both research design and methodology. Unlike traditional approaches that fit structural equation models to quantitative data collected via self-reported Likert-type scales in self-administered questionnaires, this study adopts an innovative approach. Doing so, this study goes beyond the limitations of the contemporary dominating paradigm in consumer disposal research (Raab, 2024) by providing experimental evidence to assess causality in combination with qualitative in-depths interviews that challenge subjective perceptions of waste separation behaviors. Environmental concerns, anticipated guilt, awareness of consequences, and health consciousness positively affect waste separation intentions. Conversely, perceived inconvenience, information, and infrastructure barriers negatively affect waste separation intentions (Kushwah et al., 2023). Therefore, user-friendly, interactive digital platforms and apps may encourage recycling through education and gamification (Sailer & Homner, 2020).

The theory of gamified learning (TGL) also incorporates game mechanics in waste separation initiatives (Landers et al., 2015), but its impact on aligning citizens' norms, intrinsic motivations, and reactions remains unexplored (Razali et al., 2020; Stoeva & Alriksson, 2017).

We contribute to narrowing this research gap by examining how

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gamified waste separation initiatives influence individuals' recycling engagement and alignment with recycling practices and regulations. This study's novelty involves pairing a gamified experiment within the TPB framework with qualitative interviews. By surpassing conventional studies that quantify the effects of attitude, social norms, perceived behavioral control, and behavior intentions (e.g., Raab et al., 2020; 2021a; Salem et al., 2020), this study provides evidence of the components and drivers of disposal behavior through examples and testimonials for each variable. Attitudes toward recycling are enhanced by the provision of gamified information methods, which demonstrates that behavioral intentions are strongly influenced by learning, and learning is, in turn, affected by gamification. This underscores the importance of considering information methods when local governments seek to affect people's behavior.

This article proceeds with a literature review, the research propositions and methods, the results and discussion, a review of this work's contributions, and conclusions.

The study significantly contributes to the Theory of Gamified Learning (TGL) by demonstrating how gamification can effectively enhance learning, particularly in recycling and sustainable behavior change. The findings show that gamified apps can improve understanding of waste separation practices, supporting TGL's principles of increasing engagement through game mechanics. However, the study also reveals that gamification's effectiveness varies depending on the audience and specific game mechanics, suggesting it may work better as a supplementary tool rather than a primary approach in some cases. This nuance advances the understanding of TGL and its application in educational contexts.

2. Literature review

2.1. Determinants of household solid waste separation behavior

Studies of municipal waste generation in the European Union reveal significant differences of magnitude between countries. These disparities are attributed to various factors, particularly economic development (Minelgaitė & Liobikienė, 2019). Research into the link between economic prosperity and recycling rates is limited but indicates a positive relationship (Bobba et al., 2023; Boubellouta & Kusch-Brandt, 2023). Recycling behavior also correlates with waste generation levels, implying a connection to positive attitudes toward waste reduction (Vicente & Reis, 2008). Administrative, legal, economic, and other measures seek to alter individuals' attitudes and behaviors (Bernstad, 2014; Knickmeyer, 2020; Ma et al., 2018; Raab et al., 2020, 2021a; Stoeva & Alriksson, 2017; Xu et al., 2018). Recent studies integrate the dual factor, self-determination, and status quo bias theories to explore the antecedents influencing households' waste separation intentions (Hu et al., 2024; Kushwah et al., 2023). Environmental concern, awareness of consequences, and health consciousness positively influence waste separation intentions, whereas perceived inconvenience, inadequate knowledge, hygiene concerns, and mistrust of waste systems negatively influence those intentions (Pedersen & Manhice, 2020). Recent studies have addressed the motivation and ability to recycle; for example, T. Soma et al. (2021) use the motivation opportunity ability framework to assess the experiences of households that participated in an awareness campaign intervention study.

Adopting the Theory of Planned Behavior (TPB) to study consumer disposal and waste separation behaviors suggests that attitudes, subjective norms, and perceived behavioral control (PBC) predict sustainable behavior and (food) waste behavior (Kaplan Mintz et al., 2019; Russell et al., 2017; T. Soma et al., 2021). Specifically, attitudes toward behaviors involve evaluating the likely outcomes of a specific behavior (Botetzagias et al., 2015; Salem et al., 2020); subjective norms describe whether the social environment approves or disapproves of a particular behavior and how much an individual is influenced by their societal surroundings, namely describe environmental concern, environmental

self-identity, and civic responsibility (Botetzagias et al., 2015; Raab et al., 2021b). PBC describes individuals' perceptions and ability to control the outcomes of their behavior, influencing the likelihood of successful action (Ajzen, 1991; Botetzagias et al., 2015). Therefore, efforts to improve waste management must consider these intertwined elements (T. Soma et al., 2021). These antecedents affect intentions; as most human actions are goal-directed, a person is likely to engage in pro-environmental behavior because they intend to do so (Botetzagias et al., 2015; Raab et al., 2021a). Intention, then, affects behavior (Ajzen, 1991; Russell et al., 2017), but individuals may not fully implement their intentions, resulting in an intention-behavior gap (Echegaray & Hansstein, 2017; Wang & Mangmeechai, 2021). Bridging this gap is crucial to improve recycling and reduce incineration or landfill volume. Tailored interventions are needed to account for intrinsic motivations, resistance, and varying compliance with norms (Razali et al., 2020; Stoeva & Alriksson, 2017).

This study examines the gap that occurs between intention and behavior in the specific application domain of waste separation practices. To conduct the analysis, we considered the effects of both digitalization and gamification (Fig. 1) on attitudes, subjective norms, and PBC as intentions and behaviors.

Building on the consumer behavior framework, this study integrates the TGL by introducing digitalization and gamification as moderators and analyzing the impact of gamified and non-gamified information in contrast to paper-based information on households' waste separation decisions.

2.2. Digitalization and gamification

Digitalization and sustainability represent a paradigm shift in organizations and society, although scholarly interest in their convergence is recent (Ritala, 2024). Platforms and apps addressing various user groups are vital for this transformation as they provide infrastructure for communication, coordination, and transactions (Abbate et al., 2019; Hagi & Wright, 2015). They facilitate interactions, negotiations, and information sharing that unite diverse user segments. Studies emphasize leveraging social media, mobile tech, and online games to improve civic engagement (Cheng et al., 2021), environmental activities (Wamuyu, 2018), activism (Young et al., 2017), and stakeholder communication (Trivedi et al., 2018). Gamification, in particular, has proven effective in promoting environmental awareness and behavioral change by making sustainability initiatives more interactive and rewarding. Previous research indicates that while gamification does not significantly influence behavior, users accept games, report positive experiences, and show a high intention to use technologies and apps to separate waste properly. This indicates that the learning environment and the assistance in identifying different waste types were well-received (Shaper et al., 2022). Therefore, apps that gamify recycling or energy-saving behaviors might significantly increase users' participation and long-term commitment to sustainability, and introducing points, badges, and leaderboards could enhance user engagement and reduce the gap between intentions and behaviors.

The existing literature does not investigate in-depth how official digital communication promotes public engagement, despite the impact of digitalization and the broad utilization of apps and gamification (Subhash & Cudney, 2018).

Gamification involves inserting game mechanisms into non-game contexts. An analysis of methodologically rigorous studies found that gamification had stable effects on cognitive learning outcomes but not on motivational or behavioral outcomes (Sailer & Homner, 2020). Intermediary platforms (e.g., apps and websites) effectively promote behaviors by transforming communication into interactive experiences that engage consumers and provoke cognitive, motivational, and behavioral reactions (Kim & Baek, 2018; Stocchi et al., 2021). Apps that encourage sustainable behavior embrace various purposes that can be categorized into four groups: information and education, energy

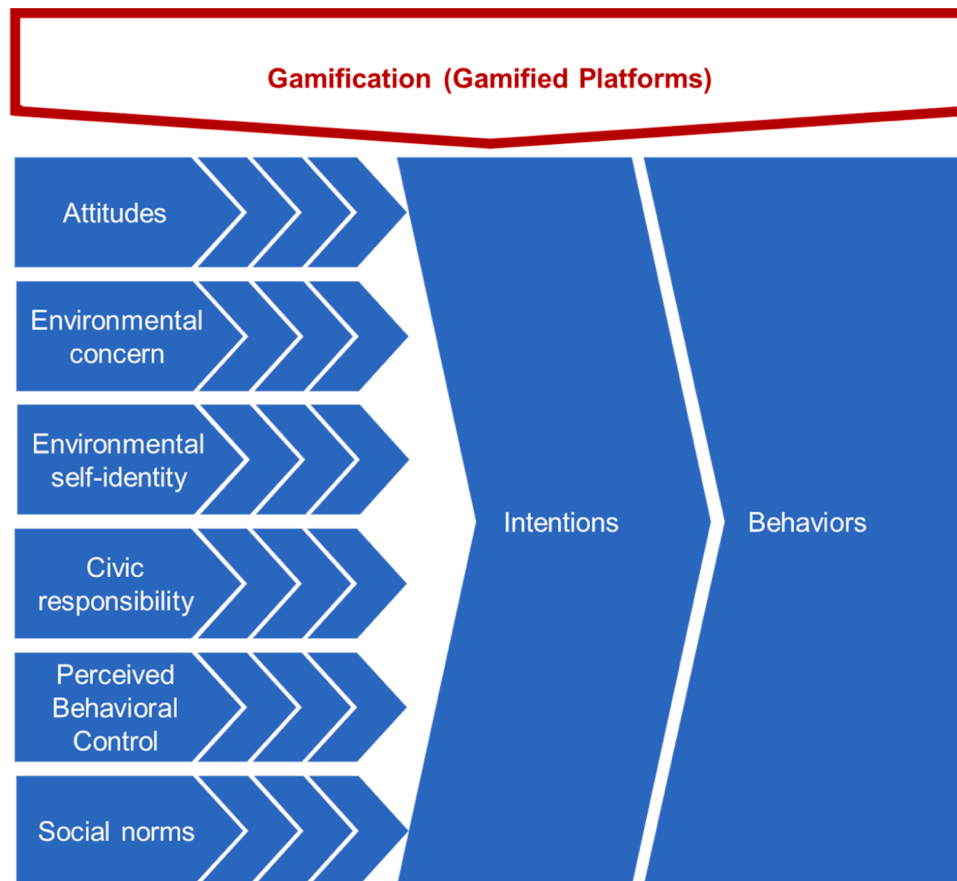


Fig. 1. Connecting the TGL with the consumer behavior framework.

reduction, transportation, and waste management (Chappin et al., 2017; Douglas and Brauer, 2021). Among others, the Donapp, Karmafy, GreenApes, Aworld, Green Karma, Clean Coin, and Junker apps are growing and recognized as key players in gamified learning (see Appendix 1).

As Hoffmann and Pfeiffer (2022) and Papamichael et al. (2022) report, games and mobile apps can affect attitudes and behaviors to encourage proper solid waste separation by informing, educating, and engaging citizens. We have formulated the following propositions to investigate the general advantages of gamified learning reported in previous research, focusing on the effects of apps, gamified apps, and conventional paper-based information on citizens' behaviors.

Proposition 1. Gamified information increases knowledge about correct recycling waste separation.

Proposition 2. General (non-gamified) information increases knowledge about correct recycling waste separation.

Proposition 3. Gamified information significantly increases the percentage of correct recycling waste separation decisions compared to non-gamified information.

Proposition 4. Uninformed citizens make a lower percentage of correct waste separation decisions than those using games.

Proposition 5. Uninformed citizens make a lower percentage of correct waste separation decisions than informed participants.

These propositions align with the research objectives of the qualitative data analysis presented in Section 4.

3. Research design and data description

This study applied a mixed-methods approach that integrated quantitative analysis with qualitative interviews. In the experiment, user-friendly waste management apps were used to engage participants

for 4 days, from November 2 to 9, 2022, and data from 18 participants were collected through questionnaires and interviews. The Junker¹ and greenApes² apps were used to teach the correct recycling modality for each product and encourage sustainable actions by offering rewards. While Junker effectively communicates about waste sorting and recycling initiatives, greenApes uses gamification to encourage sustainable lifestyles. Junker app is designed to simplify waste sorting and recycling for residents and visitors in Italy. By scanning a product's barcode, Junker identifies the item and provides detailed instructions on how to dispose of each component according to local recycling regulations. Beyond product identification, Junker utilizes geolocation to offer users personalized information based on their current location. This includes local waste collection schedules and maps of nearby recycling centers, ensuring that users have access to accurate and relevant data. greenApes engages participants into eco-friendly activities and challenges such as walking, cycling, recycling, volunteering, etc. It rewards participants positive behaviors with virtual points that can be donated to support associations and NGOs dedicated to safeguarding the planet and addressing social challenges. Participants gain access to discounts and experiences offered by greenApes' partners, including companies that provide sustainable products and services. The app also fosters a community where individuals can exchange tips and experiences and discover a wealth of content related to sustainability. As depicted in Fig. 2, a specific greenApes challenge was selected for this experiment, requiring participants to watch educational content on proper waste separation and the importance of recycling. Afterward, they completed a

¹ <https://junkerapp.it/recycling-is-easy-in-italy-just-a-blip-away-with-junker/> (retrieved December 19, 2023)

² <https://www.greenapes.com/en/> (retrieved December 19, 2023)



Fig. 2. Junker and greenApes apps.

quiz based on the content. Participants earned coins based on their quiz scores, which were assessed in collaboration with the researcher during the final scoring process.

An Italian location was selected due to Italy's prominence in recycling and achieving a circular economy, as Eurostat data from 2021 affirms.³ The distinct waste separation regulations in each town necessitated the selection of a specific city (Cerqueti et al., 2021). Pistoia in Tuscany was chosen. A group of volunteers was contacted, and 26 participants were selected. In phone calls, participants were informed about the experiment and the GDPR conditions. The study was conducted with clear and informed consent from all participants, ensuring they fully understood the purpose and scope of data usage. Ethical oversight was deemed unnecessary due to the nature of the study. All participants were reassured of the anonymity of their responses and the independence of the research from the Municipality of Pistoia and ALIA, the public company managing waste collection in Pistoia.

The volunteers were divided into four groups: three for the experiments and one for the interview validation. The 26 voluntary participants (aged >18 years) lived permanently in Pistoia and were engaged in waste disposal (Table 1). The sample represents the Italian population according to common characteristics, such as age distribution, educational level, geographical distribution between urban and suburban areas, and jobs, as reported by ISTAT, the Italian National Institute of Statistics. Potential participants were contacted randomly via email with the support of Pistoia's waste management system. A smaller group was then selected and randomly assigned to the four groups described in Section 3.2. Data were collected through online interviews that lasted 30 minutes on average and were conducted via telephone and video calls.

We randomly assigned the respondents to three treatment groups (Fig. 3). The first (G1) used the gamified greenApes app to participate in a challenge that provided information about accurate waste separation; the second (G2) received the same information via Junker, a non-

Table 1
Descriptive statistics.

Name	Age	Gender	Job	Education	Area	Group
P1	19	M	Student	Diploma	Province	1
P2	24	F	Student	Diploma	Suburb	1
P3	19	F	Student	Diploma	Center	1
P4	56	F	Teacher	PhD	Center	1
P5	66	M	Retiree	Master's	Suburb	1
P6	53	F	Teacher	Master's	Center	1
P7	56	M	Bank clerk	Diploma	Province	2
P8	24	M	Student	Bachelor's	Suburb	2
P9	66	F	Pensioner	Master's	Center	2
P10	24	F	Real estate agent	Bachelor's	Center	2
P11	58	M	Teacher	Master's	Center	2
P12	57	F	Teacher	Master's	Suburb	2
P13	21	M	Student	Diploma	Center	3
P14	66	F	Pensioner	Primary school license	Province	3
P15	80	F	Pensioner	Master's	Suburb	3
P16	73	M	Pensioner	Master's	Suburb	3
P17	59	M	Teacher	PhD	Center	3
P18	59	M	Financial advisor	Master's	Center	3
P19	39	F	Cultural freelancer	Master's	Center	4
P20	47	F	Housewife	Diploma	Suburb	4
P21	46	M	Employee - transports	Master's	Suburb	4
P22	43	M	Policeman	Bachelor's	Center	4
P23	36	F	Psychologist	Master's	Suburb	4
P24	27	M	Student	PhD	Center	4
P25	32	F	Teacher	Master's	Suburb	4
P26	31	M	Unemployed	Master's	Center	4

Diploma = high school diploma; master's = master's degree; bachelor = bachelor's degree

gamified app; the control group (G3) was not informed with multimedia inputs but only with flyers and detailed written documents provided by the municipality. As a control sample (G4), eight additional

³ https://ec.europa.eu/eurostat/databrowser/product/view/env_wasmun (retrieved December 19, 2023)

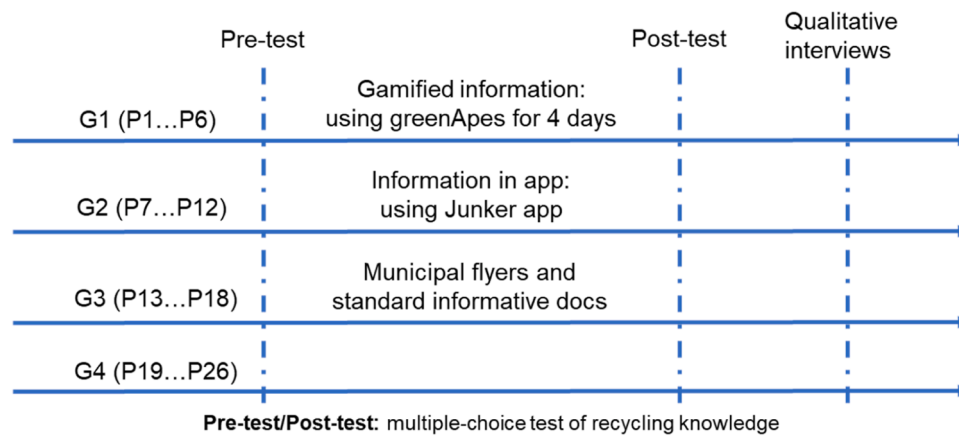


Fig. 3. Research design and treatments.

participants were interviewed to validate the results.

To assess participants' initial awareness of correct separation procedures, the participants answered seven questions about recycling rules. The post-test used altered questions to determine whether the participants exhibited improved knowledge after the treatments. Both the tests and the interview protocol were validated by the research group in collaboration with a group of volunteering students. The data were analyzed with an independent samples median test to assess significant differences in waste separation knowledge.

Semi-structured interviews were conducted in Italian via telephone or video call for approximately 30 minutes each, transcribed directly after having been conducted, and translated into English to gain insights into the participants' preferences and motivations. The 26 semi-structured interviews were analyzed with MAXQDA, and the Gioia method (Gioia, 2021) was applied. As detailed in the following sections, first-order concepts were derived directly from the text by grouping respondents' sentences according to similar topics. Second-order themes were then developed by aggregating first-order concepts based on theoretical frameworks and common categories identified from the responses. For the aggregate dimension of Motivational Factors, second-order themes were further consolidated into third-order themes, guided by theoretical insights and research propositions. Ultimately, second and third-order themes were organized into two overarching aggregate dimensions: Motivational Factors and Learning Behavior.

Data, interview transcriptions, and codes have been discussed regularly among team members to discuss and reconcile discrepancies (Boyatzis, 1998). Axial coding and second-order themes. Axial coding (Corbin and Strauss, 1990) was used to establish links between the first-order codes and to assemble them under higher-order themes.

4. Results

In this section, we present the outcomes, focusing on the experimental evaluation of the research propositions. We then discuss the resulting antecedents of waste separation behaviors. Finally, we discuss the motivational factors driving proper waste separation practices.

4.1. Evaluation of learning behavior and knowledge transfer

To assess respondents' knowledge gain (Table 1) measured as the number of correct answers about proper waste disposal in the pre and post-tests, we used non-parametric independent samples median tests because the sample size did not meet the minimum size requirements of conventional *t*-tests. This study (Table 2) revealed consistent medians in pre-treatment knowledge across groups, indicating no significant variation in knowledge about waste recycling between the groups. However, the analysis of the differences in knowledge between the pre and post-

Table 2
Knowledge test results.

Name	Group	Pre-test	Post-test
P1	1	43 %	86 %
P2	1	57 %	100 %
P3	1	29 %	86 %
P4	1	100 %	57 %
P5	1	57 %	71 %
P6	1	57 %	86 %
P7	2	43 %	86 %
P8	2	57 %	71 %
P9	2	86 %	86 %
P10	2	71 %	86 %
P11	2	71 %	57 %
P12	2	71 %	100 %
P13	3	71 %	71 %
P14	3	57 %	57 %
P15	3	86 %	86 %
P16	3	71 %	71 %
P17	3	71 %	71 %
P18	3	14 %	71 %

tests showed a rejection of the hypothesis of identical medians among groups, indicating different post-test results after the distinct treatments (Table 2).

Fig. 4 highlights the differences between the gamified and non-gamified groups' results compared to the control. The control's correct answers remained stable, while both treated groups improved.

The non-parametric independent samples median test was chosen to ensure valid statistical test results despite the relatively small sample size.

Notably, while G1 showed slightly more improvement, no significant difference was observed between that group and G2. Both gained knowledge and enhanced recycling practices compared to the control group (G3). While G3, the control group's results did not change, G1 and G2 substantially increased their knowledge; therefore, propositions 1 and 2 are supported. G1's improvement was not significantly higher than G2's, so proposition 3 is rejected. Propositions 4 and 5 are supported as participants in the treated groups significantly improved the correctness of their answers compared to the control group.

According to the qualitative analysis, depicted in Fig. 5, most interviewees gained knowledge about how to recycle through documents and internet browsing (58 %).

Participants in G2 who used the Junker app appreciated the opportunity to retrieve information on demand, improving their knowledge and informed decisions about waste disposal and recycling.

However, participants in G1 who used the gamified app had very mixed reactions. None of them had used a gamified recycling app before. The majority reported specific negative feedback about the game

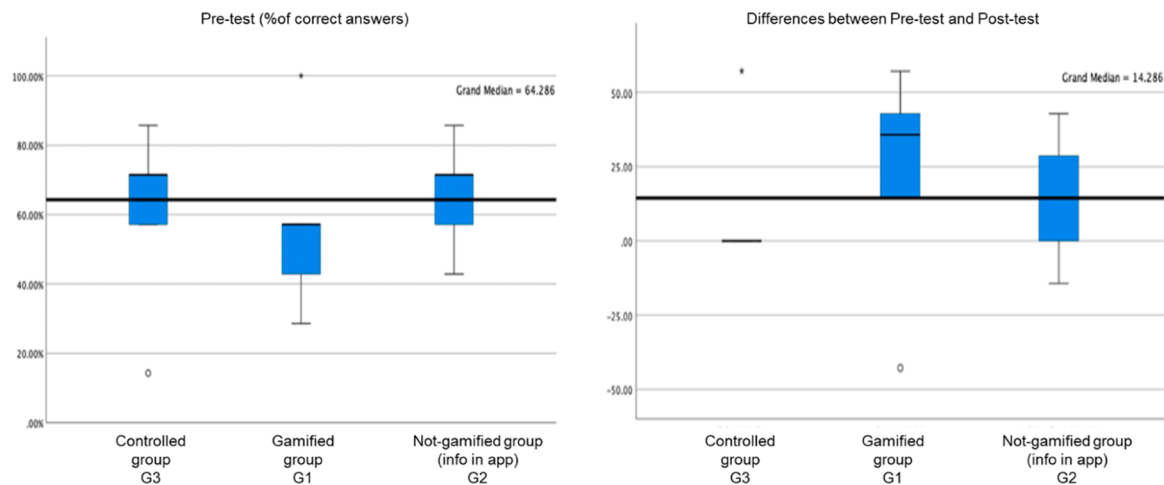


Fig. 4. Non-parametric independent samples median test results.

mechanics, e.g., “The part with the games ... motivated me the least” (P4); “I enjoyed the challenges less” (P3); and “Points and gamification ... motivated me the least” (P6). Participants in G1 also disliked specific app functions, e.g., “I didn’t like the social part” (P1) and “I liked the social part a bit less” (P2). Only three respondents described the gamified app as highly effective for acquiring proficiency in waste segregation: “I downloaded an app ... to understand better [the rules]” (P6), and “I learned, and I found it useful. ... I even promoted the app” (P4). The apps provided easy guidance: “The app motivates you” (P3) and was “more functional” (P1). Some resistance to digital adoption was found among participants older than 60 years.

Even without using the gamified app, participants expressed skepticism about gamification’s relevance to waste recycling, saying, “I don’t care at all about gamification” (P11), and “I wouldn’t do it [recycling] because of the game” (P12). Others maintained neutrality toward gamified recycling and showed interest in the prize or gratification element: “I would only do it for point collection, not for [the game] itself” (P13), and “[An app could be useful] for the gratification ... for having behaved virtuously” (P18). Those who did not use gamified apps said games are for children (P25) and waste time (P22). Notably, participants under 30 years viewed gamification favorably: “I prefer the application because it gave me a method to reason behind it” (P1); “For the target age of youth, the app really helps you more” (P2). Men generally liked gamification more than women (P18, P21, P23, P26, P27).

Overall, G1 and G2 mostly and surprisingly considered gamification and game elements a waste of time or a hindrance to learning. Only a few participants generally liked the gamified app because they perceived that they were learning, but having to do tasks to level up or accumulate points made them unwilling to play.

4.2. Antecedents of waste separation behavior

The analysis was intended to develop a comprehensive understanding of preferred methods of learning about waste separation and identify the rationale behind correct waste separation. We categorized the core theme as “motivational factors,” including the three antecedents of TPB. The essential subtopics contributing to this theme encompassed attitudes; subjective and social norms, such as environmental concern, environmental self-identity, civic responsibility, economic incentives, PBC; intentions; and engagement in recycling behaviors (Fig. 6).

All participants expressed a positive attitude toward recycling, describing it as societally positive, and 34 % contended that “separate collection is right” and “useful.” Some (12 %) believed that “separate collection is ... fundamental,” but others viewed separate collection negatively, e.g., “Separate collection is more difficult and uncertain”

(P20), and “tends to disincentivize reuse” (P17). The participants lacked information about the proper disposal of rarer items, which led them to feel frustrated and fail to comply with collection rules. Some found separate waste collection tiring and impractical as it requires dividing items into the correct bins (P11, P24). A young participant reported that older relatives struggled to sort waste due to inadequate information or skill and often did not even try (P12).

Some participants indicated they managed waste sorting effortlessly, without significant obstacles, e.g., “It takes no effort” (P3) and “doesn’t bother me” (P14). Those who addressed PBC highlighted problems that impede optimal collection; some noted that limited space at homemade separation difficult, e.g., “You need a lot of bins, where do you keep them?” (P9) and “I don’t have physical space” (P20).

Social norms are important in this context; eight participants reported learning about recycling from family, particularly their parents: “Separate collection has always been made in my house” (P2) and “I was taught in my family” (P7). Acquaintances influenced older participants who did not grow up recycling (P21). In contrast, some participants said their families paid inadequate attention to waste separation, which affected their behavior (P10, P22). The dominant negative influence was a rumor about improper municipal waste disposal that sowed doubt and mistrust and made some participants question their efforts’ efficacy (P14): “Those who manage separate collection wreck everything” (P26) and “It’s unknown if the municipality manages the waste in the correct way” (P19). These beliefs undermined incentives to sort waste correctly.

Environmental concerns were the respondents’ primary motivations for recycling, including nature, environmental impact, climate change, and resource safeguarding. They stated that they recycled “For nature and for the environment” (P5, P12, P22, P25, P26); “To limit the environmental impact” (P24, P20, P19); because “The climate has changed, and I’m scared” (P15); and “To save resources” (P1). None claimed that the environment was unimportant. Those with pro-environmental identities often felt responsible for protecting the planet: “I feel my burden on the planet, and I want to make up for it” (P17) and “I do it out of a personal predisposition” (P22). They hoped to gain advantages in the long run: “I have an indirect gain” (P1).

Most participants agreed with separate waste collection and wanted to continue, although they sometimes resorted to putting items in the undifferentiated bin due to haste or doubt (P16, P17): “Recycling collection is a must” (P18); “You never stop learning about recycling” (P9). Two pursued proper separation “to save as much waste as possible from the incinerator” (P16) and avoid unnecessary industrial processes: “To save an industrial effort” (P17).

The participants provided various reasons for their behaviors, including civic obligation or engagement: “I do what I’m told to do, in this case from the municipality” (P15). Others emphasized ethical

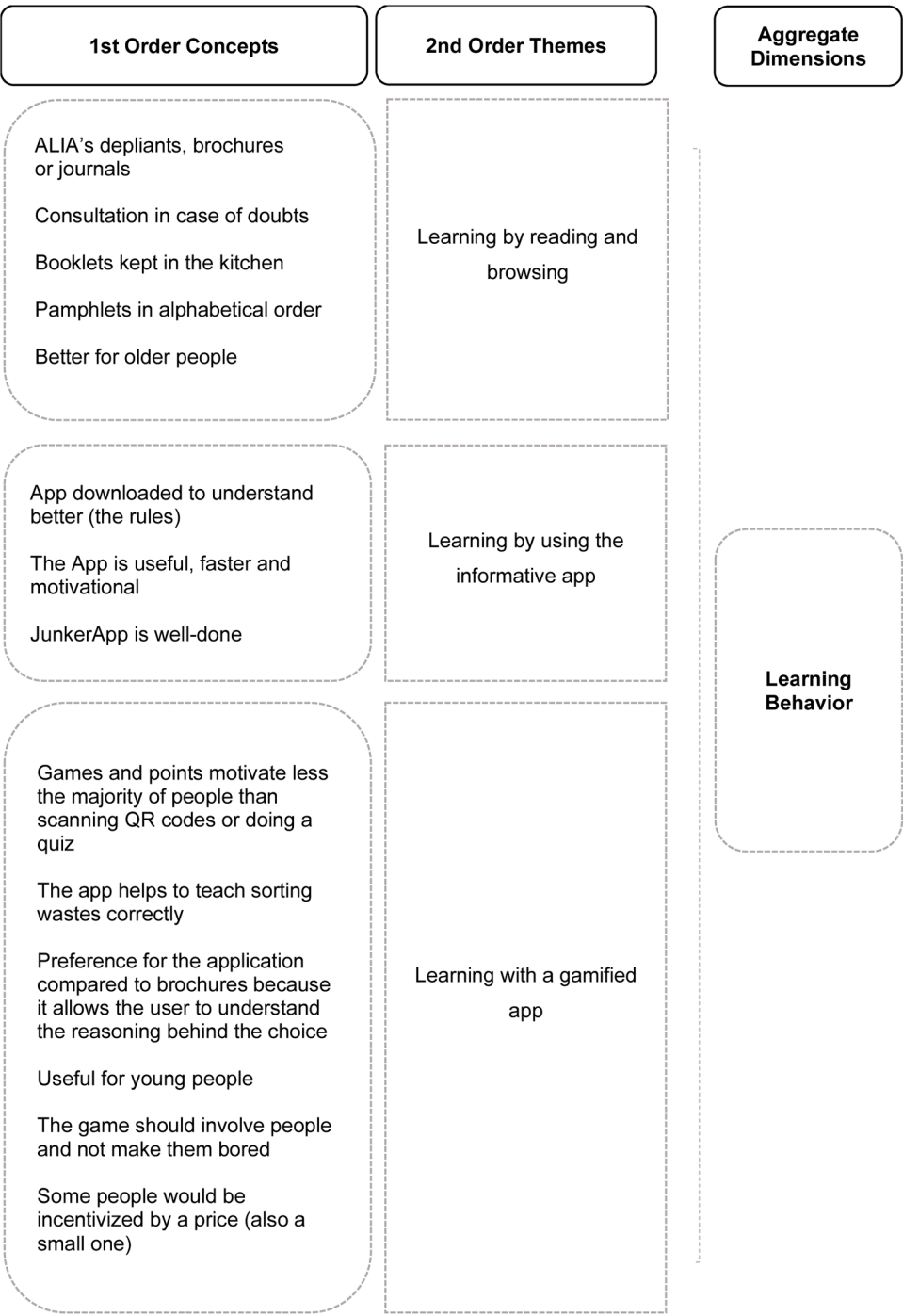


Fig. 5. Learning behaviors.

correctness (civic engagement): “It seems ethically right” (P11) and “It’s a civic duty” (P13).

Some indicated they would use a gamified recycling app only for material rewards (P10, P13, P23): “I would only do it for point collection” (P13); “If there were catchy prizes, I would use it more” (P23). The desired rewards varied; some suggested a charitable donation (P22), and others, gadgets or sustainability-related items (P21). Other participants stated that they would be more motivated to recycle by a monetary incentive, e.g., a can and bottle deposit (P5, P7, P17). Interestingly, those who preferred monetary incentives were all men over 50.

Intention represents the link between motivational factors and learning behavior. Notably, the participants highlighted their intentions to continue learning about proper waste separation. Only two

participants (P22, P26) felt satisfied with the provided information.

5. Discussion and contributions

5.1. Discussion

The results of examining the main challenge in transitioning to a circular economy—disposal decisions that determine both the quantity and quality of waste production as a resource for further processes (Knickmeyer, 2020)—indicate that gamification exploits the effects of attitude, subjective norms, and behavioral control on consumers’ correct sorting decisions.

External validity has been strengthened by carefully contextualizing

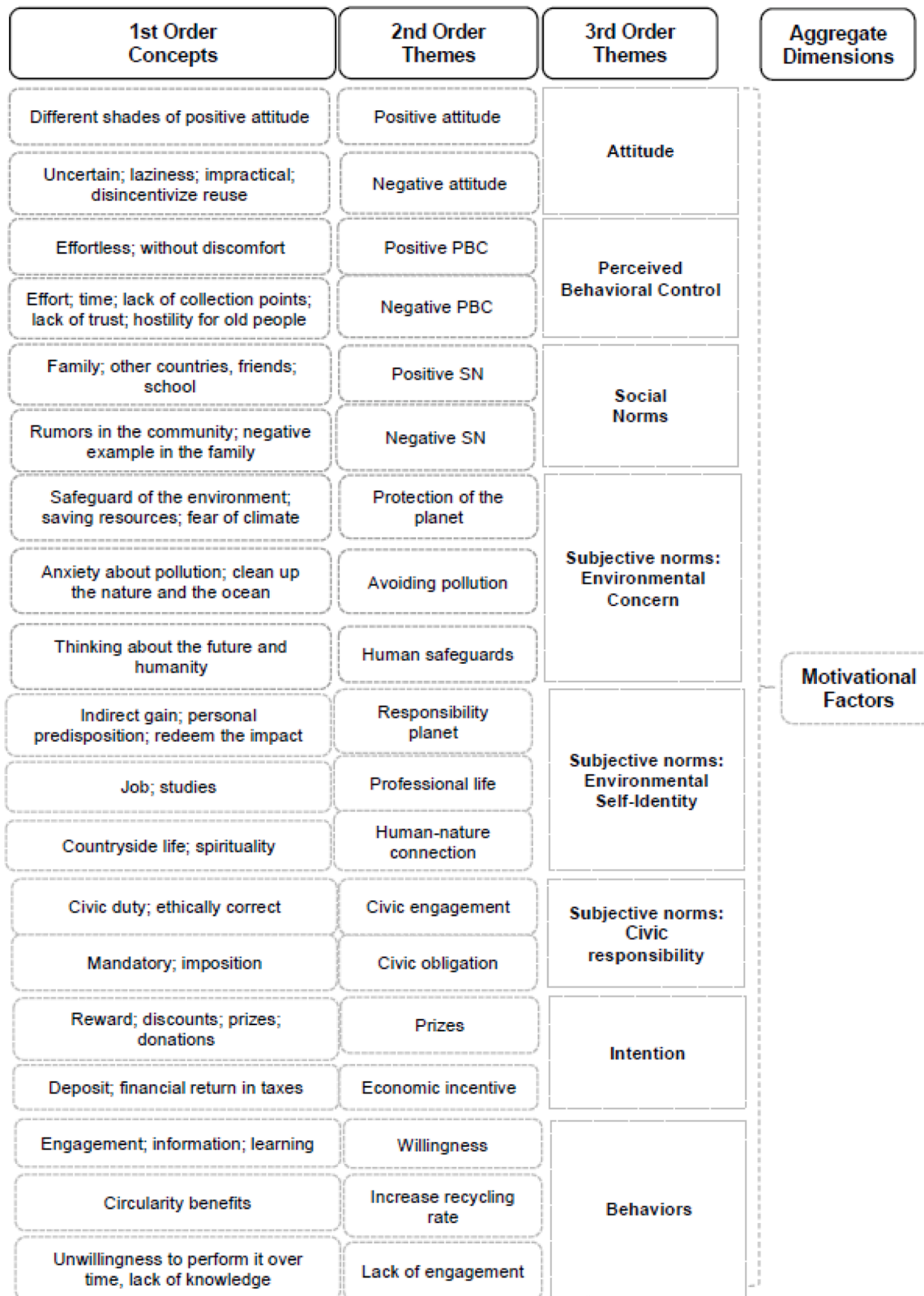


Fig. 6. Motivational factors.

the findings within the specific framework and conditions under which the research was conducted (Hayashi et al., 2019). The study took place in Pistoia, Tuscany, utilizing a diverse sample that mirrors the broader Italian population in terms of age distribution, education level, and geographic spread, as corroborated by ISTAT data. While the sample size

and geographic focus of the study might initially suggest limitations in generalizability, the insights gained are likely transferable to other contexts, particularly those with similar socio-cultural and environmental conditions. This broader applicability is supported by the theoretical foundations of the TPB and the TGL, which provide robust

frameworks for understanding behavior change across various settings. Moreover, the study’s findings on the impact of gamification and digitalization on recycling behavior are grounded in widely recognized psychological mechanisms—such as social norms, perceived behavioral control, and intrinsic motivation—that are relevant across diverse contexts. Therefore, although the findings are deeply rooted in the specific context of Pistoia, the underlying principles are broadly applicable, provided that interventions are tailored to align with local conditions and regulations. This approach is consistent with previous research on waste management practices across different municipalities (Cerqueti et al., 2021; Knickmeyer, 2020), which highlights the importance of adapting strategies to local contexts to ensure effectiveness.

Notably, the responses of the group using the gamified greenApes app varied significantly from the control group’s responses but not from those of participants using the non-gamified Junker app, who learned by browsing information. Respondents appreciated being able to retrieve information according to their actual needs; they disliked the game mechanics because they spurred them to interact with the app without a concrete need. Gamification was no more effective than textual information in motivating individuals, partly due to participants’ low level of digital adoption that limited the positive effects of digitalization.

All the experimental and interview data provided us with numerous insights. Table 3 summarizes the research results and the identified impacts of antecedents on behavior intentions and actual behavior.

To further discuss the results presented in the table, we depict some paradigmatic examples.

Negative word of mouth affects behavioral intentions, such as the rumor that household recycling is being thwarted by improper municipal disposal. Moreover, improper family disposal habits negatively affect behavioral intentions as some participants reported: “In my home, there isn’t this attitude” (P10) or “My family didn’t pay attention to this theme” (P22).

Gamification and game elements were often perceived as a waste of time or a hindrance to learning by many users. While the app GreenApes was generally liked for its educational value, the requirement to complete tasks or accumulate points to progress discouraged engagement. Some non-users associated games with children and preferred faster methods for learning about waste separation, labeling the game as

inefficient. However, participants under 30 expressed positive views about the app and gamification. These findings support previous research results (Botetzagias et al., 2015; Pedersen & Manhice, 2020). Building on the conclusions of Razali et al. (2020) and Stoeva & Alriksson (2017), which emphasize the need for tailored interventions to improve households’ waste separation behavior, this study provides evidence for the most effective mode of information delivery. In doing so, it addresses the gap in the nuanced understanding required for tailoring interventions, offering practical insights for designing more targeted and effective strategies.

Notably, the social pressure for proper waste separation is positive, as expected, but the social pressure to use the app negatively affects behavioral intentions. This reaction indicates that reactance must be considered in future studies and industrial applications to motivate, inform, or nudge citizens using apps such as greenApes or Junker.

5.2. Contributions to theory building and maintenance

The TPB, the leading explanatory framework for consumers’ disposal behaviors (Raab, 2024), frequently guides research that uses quantitative surveys for data collection and structural equation modeling for data analysis. This study demonstrates how experiments and qualitative interviews can be fruitfully used to complement each other within this framework. Both positive and negative influences on behavior are revealed with this method, which opposes Salem et al.’s (2020) findings on the impact of behavioral control on citizens’ waste reduction.

Another interesting advancement comes from the analysis of the qualitative interviews, which enrich the spectrum of antecedents by offering a broader interpretation of “subjective norms,” including the environmental concerns that are depicted in second-order themes as the reflected self-identity (acting responsibly to protect the planet, handling professional life and human–nature connections) and civic responsibility focusing on civic engagement and obligations (see Table 3). As these antecedents would not have been identified by using the TPB framework in the conventional quantitative manner, this study demonstrated the value that qualitative interviews added to the TPB framework.

Furthermore, this study makes substantial contributions to the Theory of Gamified Learning (TGL) by demonstrating the effective application of gamification in educational contexts, particularly in promoting recycling and sustainable behavior change. The findings show that gamified apps, such as those used in the experiment, can significantly enhance the learning and comprehension of correct waste separation practices. This aligns with the principles of TGL, which suggest that integrating game mechanics into non-game contexts can boost user engagement and motivation to learn, as emphasized by Schaper et al. (2022).

However, the study also reveals that gamification is not universally more effective than traditional, non-gamified approaches. This insight broadens the understanding of TGL by illustrating that the success of gamification may vary depending on the target audience and the specific game mechanics employed. It suggests that in some cases, gamification may be more effective as a supplementary tool rather than as the primary method in the learning process—a nuance also highlighted by Sailer and Homner (2020). In summary, this study significantly advances the Theory of Gamified Learning by providing empirical evidence of how gamification can be effectively applied in educational settings, especially in the context of waste-sorting and fostering sustainable behavior change. On the other hand, this underscores the need for further research on gamification and learning to develop a comprehensive model that integrates and evaluates diverse forms of gamification, technological tools, and educational materials in waste management.

Table 3
Synopsis of results.

Using Apps (Gamified and Non-Gamified) Gamification greenApes; game mechanics and specific features (+−) Junker; easily guides (+) resistance to digitalization (−)		
Attitudes	Intentions	Behavior
- recycling collection (+) - separation is tiring (−) - more information needed (−)	- environmental concern (+) - pro-environmental identity (+) - engagement (+) - civic responsibility (+−) - economic motivation (+−)	- learning: documents and browsing the internet - learning with an app: employing the non-gamified app as an effective tool (+) - learning with a gamified app: waste of time (−) hindrance to learning (−)
Subjective norms - social pressure with app (−) - social pressure for proper waste separation (+) - negative word of mouth, improper disposal (−)		
PBC - effortless; no obstacles (+) - problems with recycling collection; space (−) - doubts and less trust concerning the waste company (−)		

5.3. Practical contributions

Our research provides practical insights for municipalities and app developers. Distinct approaches must be considered to optimize waste separation through apps. Non-gamified apps are effective at enhancing awareness and knowledge dissemination. To integrate gamification, concerns about game elements detracting from learning objectives must be addressed. Balancing engagement with educational content is crucial. Clear guidance in the in-app interface design also enhances learning and addressing negative social aspects within apps improve user engagement and efficacy in waste segregation. At the municipal level, enhancing the dissemination of recycling information is crucial (e.g., through programs that allow participants to follow their waste virtually to the recycling station). Comprehensive information about recycling methods alleviates confusion. Collaborative efforts between municipalities and app developers can enhance waste management practices by aligning strategies to close information gaps and provide user-centric designs. Fostering trust in waste management companies through transparent communication and proactive measures is vital, as is countering negative perceptions via transparent communication to improve public engagement.

6. Conclusions and limitations

This study demonstrates that using gamified and non-gamified apps effectively improved participants' knowledge and learning attitudes toward waste separation accuracy over the control group. Although it highlighted the importance of informing consumers about recycling, the gamified approach did not produce significantly better results compared to the non-gamified approach as participants learned recycling techniques from browsing information in the non-gamified app. In contrast with the common expectation that gamification and digitalization are transformative in promoting more sustainable behaviors (Ritala, 2024), our findings reveal that they are not inherently superior in changing household waste separation behavior.

Gamification encouraged positive attitudes toward waste separation by making learning engagement, but user preferences varied. Environmental concerns drove participants' intentions, while social norms significantly influenced their recycling behavior. We also expanded the TPB by identifying additional antecedents of subjective norms.

The limitations of this study arise from its small sample size. However, rather than focusing on statistical generalization, qualitative research often aims for analytical generalization. This approach involves examining whether the theoretical concepts developed from the data can be applied to other contexts or situations. The method of "analytical generalization" is based on this principle. It seeks to generalize or extend the significance of specific empirical patterns observed in a particular case or small sample by applying theoretical concepts. In essence, it involves inferring broader theories or concepts from the unique characteristics of a case, with the idea that these could also apply to other, similar cases (Delmar, 2010; Halkier, 2011).

Future research should further investigate gamification's influence across groups, (e.g., age, education, environmental identity, and propensity to play), devise strategies to counter resistance, and refine interventions through in-depth experiments. For example, connecting gamified app users' scoreboards with social media might increase their motivation to engage with the gamified app and simultaneously maintain public awareness of the need to separate waste properly.

However, the applicability of the app across different municipalities requires adjustments to align with the specific waste separation regulations of each locality. Such adaptations are essential to ensure that the app complies with the unique requirements and policies of individual municipalities, which result in heterogeneous waste management practices (Cerqueti et al., 2021). Despite these necessary modifications, there is no compelling reason to believe that the differences observed among the behavioral groups identified in our study would manifest

differently in other municipalities. The fundamental principles of behavior change and gamification explored in our research are broadly applicable and are grounded in widely recognized psychological mechanisms that are likely to operate similarly across various contexts. Therefore, we suggest that the findings are generalizable to other municipalities, provided the app is appropriately tailored to meet local regulatory conditions.

CRediT authorship contribution statement

Silvia Venturi: Methodology, Investigation, Data curation. **Katrin Zulauf:** Writing – original draft, Supervision, Methodology, Conceptualization. **Roberta Cuel:** Writing – review & editing, Writing – original draft, Visualization, Methodology. **Ralf Wagner:** Writing – review & editing, Writing – original draft, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.resconrec.2024.108108](https://doi.org/10.1016/j.resconrec.2024.108108).

Data availability

Data will be made available on request.

References

- Abbate, T., Codini, A.P., Aquilani, B., 2019. Knowledge co-creation in open innovation digital platforms: processes, tools and services. *J. Bus. Ind. Mark.* 34 (7), 1434–1447.
- Ajzen, I., 1991. The theory of planned behavior. *Organ. Behav. Hum. Decis. Process.* 50 (2), 179–211.
- Bernstad, A., 2014. Household food waste separation behavior and the importance of convenience. *Waste Manag.* 34 (7), 1317–1323.
- Bobba, S., Eynard, U., Maury, T., Ardente, F., Blengini, G.A., Mathieux, F., 2023. Circular Input Rate: novel indicator to assess circularity performances of materials in a sector—Application to rare earth elements in e-vehicles motors. *Resour. Conserv. Recycl.* 197, 107037.
- Botetzagias, I., Dima, A.F., Malesios, C., 2015. Extending the theory of planned behavior in the context of recycling: The role of moral norms and of demographic predictors. *Resour. Conserv. Recycl.* 95, 58–67.
- Boubellouta, B., Kusch-Brandt, S., 2023. Driving factors of e-waste recycling rate in 30 European countries: new evidence using a panel quantile regression of the EKC hypothesis coupled with the STIRPAT model. *Environ. Dev. Sustain.* 25 (8), 7533–7560.
- Boyatzis, R.E., 1998. Transforming qualitative information. thematic analysis and code development. Sage, Thousand Oaks, CA.
- Cerqueti, R., Cinelli, M., Minervini, L., 2021. Municipal waste management: A complex network approach with an application to Italy. *Waste Manage.* 126, 597–607. <https://doi.org/10.1016/j.wasman.2021.03.035>.
- Chan, R.B.Y., 2022. Packaging solutions for household food waste in the context of the food/beverage-packaging industry: A comparative review of empirical literature and industry press releases. *Resour. Conserv. Recycl.* 185, 106479.
- Chappin, E.J., Bijvoet, X., Oei, A., 2017. Teaching sustainability to a broad audience through an entertainment game—The effect of Catan: Oil Springs. *J. Cleaner Product.* 156, 556–568.
- Chen, L., Gao, M., 2021. Novel information interaction rule for municipal household waste classification behavior based on an evolving scale-free network. *Resour. Conserv. Recycl.* 168, 105445.
- Cheng, K.M., Koo, A.C., Mohd Nasir, J.S., Wong, S.Y., 2021. Playing Edcraft at Home: Gamified online learning for recycling intention during lockdown. *F1000Research* 10 (890), 1–17. <https://doi.org/10.12688/f1000research.72761.2>. ISSN 2046-1402.
- Corbin, J., Strauss, A., 1990. Grounded theory research: procedures, canons and evaluative criteria. *Qual. Sociol.* 13, 3–21. <https://doi.org/10.1007/bf00988593>.
- Delmar, C., 2010. Generalizability as recognition: reflections on a foundational problem in qualitative research. *Qualitative studies* 1 (2), 115–128.
- Douglas, B.D., Brauer, M., 2021. Gamification to prevent climate change: a review of games and apps for sustainability. *Curr. Opin. Psychol.* 42, 89–94.
- Echegaray, F., Hansstein, F.V., 2017. Assessing the intention-behavior gap in electronic waste recycling: the case of Brazil. *J. Clean. Prod.* 142, 180–190.

- Gioia, D., 2021. A systematic methodology for doing qualitative research. *J. Appl. Behav. Sci.* 57 (1), 20–29.
- Hagiu, A., Wright, J., 2015. Multi-sided platforms. *Int. J. Ind. Organ.* 43, 162–174.
- Halkier, B., 2011. Methodological practicalities in analytical generalization. *Qualitat. Inquiry* 17, 787–797. <https://doi.org/10.1177/1077800411423194>.
- Hayashi, P., Abib, G., Hoppen, N., 2019. Validity in qualitative research: A processual approach. *Qual. Rep.* 24 (1), 98–112.
- Hoffmann, G., Pfeiffer, J., 2022. Gameful learning for a more sustainable world: Measuring the effect of design elements on long-term learning outcomes in correct waste sorting. *Bus. Inf. Syst. Eng.* 1–24.
- Hu, J., Miao, L., Han, J., Zhou, W., Qian, X., 2024. Waste separation behavior with a new plastic category for the plastic resource circulation: Survey in Kansai. *J. Environ. Manag.* 349, 119370.
- Kaplan Mintz, K., Henn, L., Park, J., Kurman, J., 2019. What predicts household waste management behaviors? Culture and type of behavior as moderators. *Resour. Conserv. Recyc.* 145, 11–18.
- Kim, S., Baek, T.H., 2018. Examining the antecedents and consequences of mobile app engagement. *Telemat. Inform.* 35 (1), 148–158.
- Knickmeyer, D., 2020. Social factors influencing household waste separation: A literature review on good practices to improve the recycling performance of urban areas. *J. Clean. Prod.* 245, 118605.
- Kushwah, S., Gokarn, S., Ahmad, E., Pant, K.K., 2023. An empirical investigation of household's waste separation intention: A dual-factor theory perspective. *J. Environ. Manag.* 329, 117109.
- Landers, R.N., Bauer, K.N., Callan, R.C., Armstrong, M.B., 2015. Psychological theory and the gamification of learning. In: Reiners, T., Wood, L. (eds) *Gamification in education and business*. Springer, Cham. 165–186, https://doi.org/10.1007/978-3-319-10208-5_9.
- Liu, A., Mukheibir, P., 2018. Digital metering feedback and changes in water consumption—A review. *Resour. Conserv. Recyc.* 134, 136–148.
- Ma, J., Hipel, K.W., Hanson, M.L., Cai, X., Liu, Y., 2018. An analysis of influencing factors on municipal solid waste source-separated collection behavior in Guilin, China by using the theory of planned behavior. *Sustain. Cities Soc.* 37, 336–343.
- Minelgaitė, A., Liobikienė, G., 2019. Waste problem in European Union and its influence on waste management behaviours. *Sci. Total Environ.* 667, 86–93.
- Papamichael, I., Pappas, G., Siegel, J.E., Zorpas, A.A., 2022. Unified waste metrics: A gamified tool in next-generation strategic planning. *Sci. Total Environ.* 833, 154835.
- Pedersen, J.T.S., Manhice, H., 2020. The hidden dynamics of household waste separation: An anthropological analysis of user commitment, barriers, and the gaps between a waste system and its users. *J. Clean. Prod.* 242, 116285.
- Raab, K., 2024. A literature review on solid waste management and disposal behavior at the base of the pyramid. *Manag. Dyn. Knowl. Economy.* 12 (1), 1–20.
- Raab, K., Wagner, R., Salem, M., 2020. Feeling the waste? Evidence from consumers' living in Gaza Strip camps. *J. Consum. Mark.* 37 (7), 921–931.
- Raab, K., Salem, M., Wagner, R., 2021a. Antecedents of daily disposal routines in the Gaza Strip refugee camps. *Resour. Conserv. Recyc.* 168, 105427.
- Raab, K., Tolotti, G., Wagner, R., 2021b. Challenges in solid waste management: insights into the disposal behavior of suburban consumers in Guatemala City. *Front. Sustain. Cities* 3, 683576.
- Razali, F., Daud, D., Weng-Wai, C., Jiram, W.R.A., 2020. Waste separation at source behaviour among Malaysian households: The Theory of Planned Behaviour with moral norm. *J. Clean. Prod.* 271, 122025.
- Ritala, P., 2024. Grand challenges and platform ecosystems: Scaling solutions for wicked ecological and societal problems. *J. Prod. Innov. Manag.* 41 (2), 168–183.
- Russell, S.V., Young, C.W., Unsworth, K.L., Robinson, C., 2017. Bringing habits and emotions into food waste behaviour. *Resour. Conserv. Recyc.* 125, 107–114.
- Sailer, M., Hommer, L., 2020. The gamification of learning: A meta-analysis. *Educ. Psychol. Rev.* 32 (1), 77–112. <https://doi.org/10.1007/s10648-019-09498-w>.
- Salem, M., Raab, K., Wagner, R., 2020. Solid waste management: The disposal behavior of poor people living in Gaza. *Resour. Conserv. Recyc.* 153, 104550.
- Soma, T., Li, B., Maclaren, V., 2021a. An evaluation of a consumer food waste awareness campaign using the motivation opportunity ability framework. *Resour. Conserv. Recyc.* 168, 105313.
- Schaper, P., Riedmann, A., Oberdörfer, S., Krähe, M., Lugin, B., 2022. Addressing waste separation with a persuasive augmented reality app. *Proc. ACM Hum.-Comput. Interact.* 6 (MHCI), 1–16. <https://doi.org/10.1145/3546740>.
- Soma, T., Li, B., Maclaren, V., 2021b. An evaluation of a consumer food waste awareness campaign using the motivation opportunity ability framework. *Resour. Conserv. Recyc.* 168, 105313.
- Stocchi, L., Pourazad, N., Michaelidou, N., Tanusondjaja, A., Harrigan, P., 2021. Marketing research on Mobile apps: past, present and future. *J. Acad. Mark. Sci.* 1–31.
- Stoeva, K., Alriksson, S., 2017. Influence of recycling programmes on waste separation behaviour. *Waste Manage.* 68, 732–741.
- Subhash, S., Cudney, E.A., 2018. Gamified learning in higher education: A systematic review of the literature. *Comput. Hum. Behav.* 87, 192–20.
- Trivedi, R.H., Patel, J.D., Acharya, N., 2018. Causality analysis of media influence on environmental attitude, intention and behaviors leading to green purchasing. *J. Clean. Prod.* 196, 11–22.
- Vicente, P., Reis, E., 2008. Factors influencing households' participation in recycling. *Waste Manag. Res.* 26 (2), 140–146.
- Wamuyu, P.K., 2018. Leveraging Web 2.0 technologies to foster collective civic environmental initiatives among low-income urban communities. *Comput. Hum. Behav.* 85, 1–14.
- Wang, H., Mangmeechai, A., 2021. Understanding the gap between environmental intention and pro-environmental behavior towards the waste sorting and management policy of China. *Int. J. Environ. Res. Public Health* 18 (2), 757.
- Xu, L., Ling, M., Wu, Y., 2018. Economic incentive and social influence to overcome household waste separation dilemma: A field intervention study. *Waste Manage.* 77, 522–531.
- Young, W., Russell, S.V., Robinson, C.A., Barkemeyer, R., 2017. Can social media be a tool for reducing consumers' food waste? A behaviour change experiment by a UK retailer. *Resour. Conserv. Recyc.* 117, 195–203.
- Zhang, Q., Li, Y., Ding, L., Jiang, B., 2023. Systematic exploring domestic waste sorting behavior and influence mechanism of international students in China: based on the PKIC analysis framework. *Environ. Dev. Sustain.* 1–29.